

Fig. 1: Bundled-data handshaking

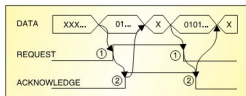


Fig. 2: Two-phase bundled-data protocol

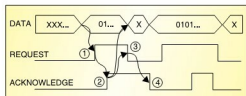


Fig. 3: Four-phase bundled-data protocol

runs at an optimal speed for the current workload. This means that it stops working if there is nothing to do.

Latest synchronous systems can also be adapted to the current workload by adjusting their clock frequency and core voltage, which is controlled by the operation system. This is not optimal because, practically, it never reaches the point where clock frequency, core voltage and workload are in an optimal balance.

Brief history of asynchronous processors

The field of asynchronous design is both old and new. ILLIAC I and ILLIAC II, which were built at the University of Illinois and became operational in 1952 and 1962, respectively, are said to have contained both synchronous and asynchronous parts. The PDP6 (Programmed Data Processor-6) from Digital Equipment (DEC) in 1960 was an asynchronous computer model. Even though asynchronous logic never disappeared completely, when clocked techniques offered an easy way of dealing with timing and hiding hazards, clockless logic was all but forgotten

until the arrival of VLSI in the late 1970s. The first Caltech Conference on VLSI in 1979 had a complete session on self-timed logic, as asynchronous logic was called at the time, within an important paper by Stucki and Cox on 'synchronisation strategies' presenting

the first pausable clock (an important device in GALS (globally asynchronous and locally synchronous)) and discussing metastability.

The first modern synthesis methods appeared around the mid 1980 with the Caltech program-transformation approach and T.A. Chu's state-transition-graph (STG) approach. Soon after, Burns and Martin, Brunvand and van Berkel proposed similar methods for the syntax-directed compilation of high-level description into asynchronous circuits.

The Petri is a more recent tool

for synthesis of asynchronous controllers described as Petri nets.

The first single-chip asynchronous microprocessor was designed at Caltech in 1988. It was followed by the first family of asynchronous clones of the ARM processor (Amulet) from University of Manchester in 1993, an 8-bit microprocessor (TITAC) from the Tokyo Institute of Technology in 1994, and the Amulet2e and TITAC-2 (a 32-bit microprocessor) in 1997. Also in 1997, the Caltech group designed the MiniMIPS—an asynchronous version of the 32-bit MIPS R3000 microprocessor. With a performance close to four times that of a clocked version in the same technology for the first prototype, the MiniMIPS is still the fastest complete asynchronous processor ever fabricated.

A group at Grenoble ported the Caltech MiniMIPS building blocks to standard cells for use in a 16-bit RISC. Other asynchronous chip experiments include the design of a fast divider at Stanford in 1991, and an instruction-length decoder for the Pentium by a research group at Intel in 1999. Low-power asynchronous microcontrollers

have been designed at Philips and Co. and Caltech Inc.

Asynchronous approach from the ground up

The main difference between synchronous and asynchronous processors is the designing technique of synchronising the sequential elements among each other. The function of the clock is to define points in time where all elements have valid and stable data at their interfaces. Between two clock events, the output signal of an element may make several transitions and none of them has to be valid as they are not relayed to their successors. In asynchronous circuits, a different technique (called 'handshaking') ensures that only valid signals are propagated.

Handshaking vs central clock.

Handshaking is one of the most important concepts of asynchronous design, as it enables the components of an asynchronous circuit to synchronise with each other for transferring data reliably and quickly without the need of taking care of the slowest link in the chain (i.e. there is no critical path which dictates minimum cycle time). Though being independent of a central clock has many advantages, it also implies difficulties that have to be dealt with, like intermediate propagation of invalid signals.

Bundled-data handshaking protocols. These protocols are common in asynchronous processors and also used in the AMULET3 asynchronous processor. They are called single-rail though bundled-data is used to describe the simultaneous transmission of control and data signals, whereas single-rail describes usage of one wire for each data bit. Now let's take a look at the different kinds of bundled-data protocols.

Two-phase bundled-data protocol. The two-phase bundled-data protocol uses a regular data path to transfer data and two additional state signals for data-send request and data-receive acknowledgment. It is used in the AMULET3 processor and often referred to as